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Week 12

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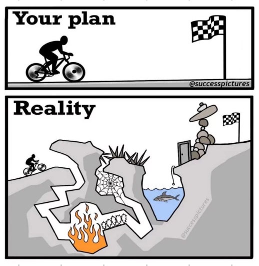
Bounded Additive Noise



↳ "Assumed System" $x(k+1) = g(x(k), u(k))$
↳ Every state is predictable

↳ "Real System" $x(k+1) = g(x(k), u(k), w(k); \theta)$
↳ Random time-dependent noises $w(t)$
↳ Unknow constants θ impacting system

↳ We normally use Additive Bounded Noise $w \in \mathcal{W}$
↳ For example wind on earth for an airplane, we know it's rarely 0, but we also know wind speed over 50 km/h usually never appears. So our $w \in \mathcal{W} \rightsquigarrow w \in (0, 50)$
↳ This noise is persistent ⇝ It never converges to 0!
↳ We will use this formulation of $w(k)$ as it models most of the world's uncertainties

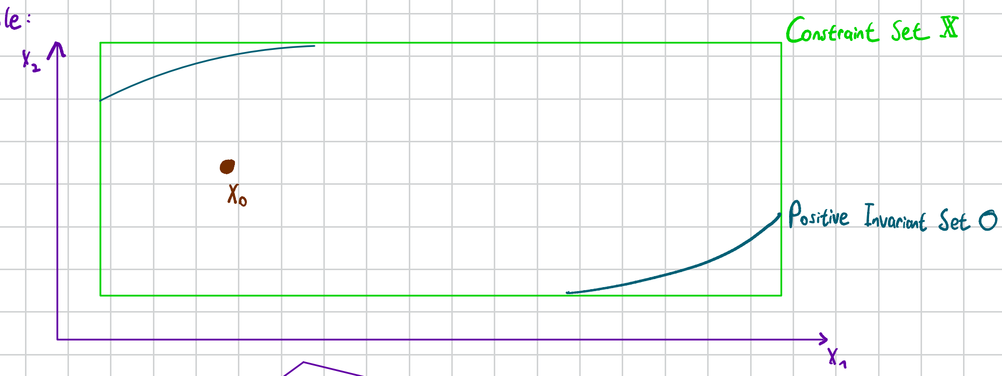
Robust Invariance

↳ Recap: Positive Invariant Set
↳ Set $\mathcal{O}$ for system $x(k+1) = g(x(k))$ is positive invariant if $x(k) \in \mathcal{O} \rightarrow x(k+1) \in \mathcal{O}$ for $k \in (0, 1, \dots)$

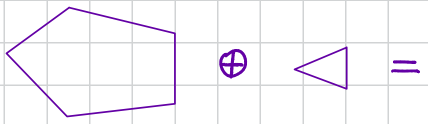
↳ New: Robust Positive Invariant Set (given some Noise $w(t)$)
↳ Set $\mathcal{O}^{\mathcal{W}}$ for system $x(k+1) = g(x(k), w(k))$ is robust positive invariant if $x \in \mathcal{O}^{\mathcal{W}} \rightarrow x(k+1) \in \mathcal{O}^{\mathcal{W}}$ for all $w \in \mathcal{W}$, $k \in (0, 1, \dots)$

↳ We can imagine, $\mathcal{O}^{\mathcal{W}} \leq \mathcal{O}$, as noises will influence our system

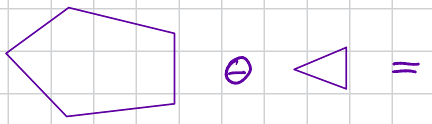
↳ Example:



↳ Minkowsky Sum $\oplus$



↳ Pontryagin Difference $\ominus$



Open Loop Prediction

↳ As we see above, we always need to take the worst possible outcome in consideration
↳ We can "simply" reformulate the problem into a nominal MPC problem simply with tighter constraints!

Nominal system	Uncertain system
$x(k+1) = Ax(k) + Bu(k)$	$x(k+1) = Ax(k) + Bu(k) + w(k)\ ,\ w \in \mathcal{W}$
$x_1 = Ax_0 + Bu_0$	$\phi_1 = Ax_0 + Bu_0 + w_0$
$x_2 = A^2x_0 + ABu_0 + Bu_1$	$\phi_2 = A^2x_0 + ABu_0 + Bu_1 + Aw_0 + w_1$
⋮	⋮
$x_i = A^ix_0 + \sum_{j=0}^{i-1} A^jBu_{i-1-j}$	$\phi_i = A^ix_0 + \sum_{j=0}^{i-1} A^jBu_{i-1-j} + \sum_{j=0}^{i-1} A^jw_{i-1-j}$
	$\phi_i = x_i + \sum_{j=0}^{i-1} A^jw_{i-1-j}$

↳ Over time, the effect of $w(k)$ will be sumed up
↳ But luckily, our nominal system has not changed

$$\phi_i(x_0, U, W) = \left\{ x_i + \sum_{j=0}^{i-1} A^jw_{i-1-j} \;\middle|\; W \in \mathcal{W}^i \right\} \subseteq \mathcal{X}$$

↳ If we assume $\mathcal{X} = \{x \mid \underbrace{Fx}_{\text{constraint}} \leq f\}$

$$Fx_i + F\sum_{j=0}^{i-1} A^jw_{i-1-j} \leq f \ \forall W \in \mathcal{W}^i$$

$$Fx_i \leq f - \max_{W \in \mathcal{W}^i} F\sum_{j=0}^{i-1} A^jw_{i-1-j}$$

↳ This part describes the Maximum effect of disturbances over time:
↳ What if the system is unstable?
↳

↳ We see, due to the open loop nature of the model, our constraint set is very conservative
↳ We might always have a strategy, but it will certainly be "too" cautious
↳ Can we do any better?

Closed Loop Prediction

↳ So let's say our model is still $x(k+1) = Ax(k) + Bu(k) + w(k)$
↳ We want to optimize over a sequence of functions $\{u_0, \mu_1, \mu_2, \dots\}$, where $\mu_i(x_i)$ is the control policy ⇝ Basically the control action at x_i
↳ As x_i is a function of w_i, every μ_i will take w_i into consideration
↳ Now, how do we find $\mu_i(x_i)$, seperation of the controll input

Pre-stabilization $\mu_i(x_i) = Kx_i + v_i$
- Fixed K, such that $A + BK$ is stable
- Simple, often conservative

The "nominal part", if there was no noise!

Linear feedback $\mu_i(x_i) = K_ix_i + v_i$
- Optimize over K_i and v_i
- Non-convex. Extremely difficult to solve...

Disturbance feedback $\mu_i(x_i) = \sum_{j=0}^{i-1} M_{ij}w_j + v_i$
- Optimize over M_{ij} and v_i
- Equivalent to linear feedback, but convex!
- Can be very effective, but computationally intense.

The "noise part", which makes sure the system still tracks the nominal state

Tube MPC $\mu_i(x_i) = v_i + K(x_i - \bar{x}_i)$
- Fixed K, such that $A + BK$ is stable
- Optimize over $\bar{x}_i$ and v_i
- Simple, and can be effective

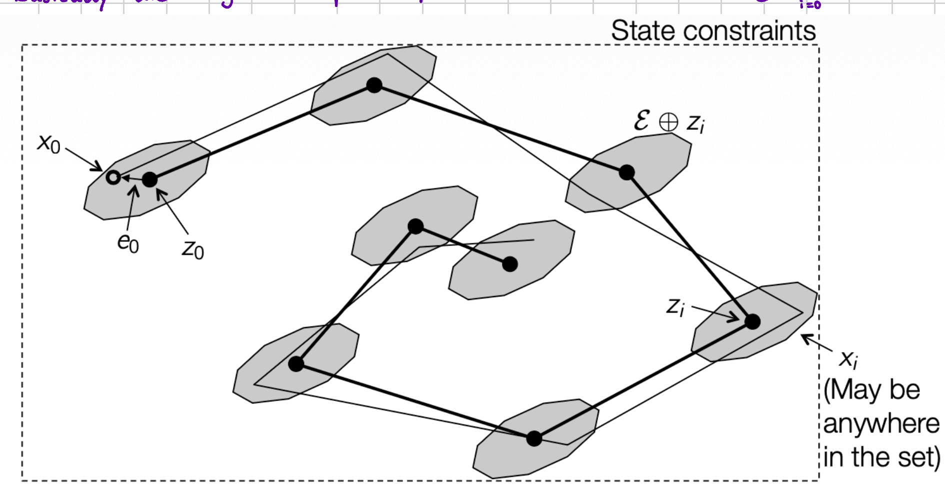
Tube MPC

↳ The big idea: $\mu_i(x_i) = v_i + K(x_i - z_i)$

We have a fixed controller K, which ensures that the system $A + BK$ stays stable

This is the nominal controller, responsible for the overall trajectory following of the states

↳ This part is also known as the error, $e_i = x_i - z_i$, and it evolves with $e_{i+1} = A_K e_i + w_i$, $A_K = A + BK$
↳ So $e_i = \sum_{j=0}^{i-1} A_K^j w_{i-1-j}$, as A_K is stable, the sum will converge to a finite number, and this is basically the "Largest error possible", or minimal Robust Invariant Set $\mathcal{E} = \bigoplus_{i=0}^{\infty} A_K^i \mathcal{W}$



↳ Based on this, our new problem formulation is now

$$J^{\star,\text{tube}}(x(k)) = \min_{V,Z} \sum_{i=0}^{N-1} l(z_i, v_i) + l_f(z_N)$$

$$\begin{aligned} \text{subj. to}\quad & z_{i+1} = Az_i + Bv_i && i \in [0, N-1] \\ & z_i \in \mathcal{X} \ominus \mathcal{E} && i \in [0, N-1] \\ & v_i \in \mathcal{U} \ominus K\mathcal{E} && i \in [0, N-1] \\ & z_N \in \mathcal{X}_f \\ & x(k) \in z_0 \oplus \mathcal{E} \end{aligned}$$

Control law: $\kappa_{\text{tube}}(x) := K(x - z_0^\star(x)) + v_0^\star(x)$

↳ Exercise for today

Exercise 2 **Tube-based MPC**

Consider the scalar linear discrete time system

$$x(k+1) = 1.1x(k) + u(k) + w(k),$$

where the state x and the input u are constrained as follows

$$\mathcal{X} = [-2, 2] \qquad \mathcal{U} = [-0.8, 0.8]$$

and the disturbance $w \in \mathcal{W} = [-0.2, 0.1]$. Design a tube-based MPC controller for this system.

1. Compute the minimal robust invariant set $\mathcal{E}$ for this system with the control law $u(k) = Kx(k)$, for $K = -0.3$

Hint : $[a, b] \oplus [c, d] = [a + c, b + d]$ and $\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}$ if $|x| < 1$